

HOSPITAL DISCHARGE READINESS INTELLIGENCE ENGINE

An AI-Based Healthcare Operations Support System

PROJECT PROBLEM-BASED DOCUMENT

Domain: Healthcare Operations

Technologies: Python, SQL, NLP, Machine Learning, SHAP, Gemini API, Streamlit

Dataset: MIMIC-IV / MIMIC-IV-Note

GitHub Link: [Hospital-Discharge-Readiness-Intelligence-Engine](https://github.com/yourusername/Hospital-Discharge-Readiness-Intelligence-Engine)

1 Problem Statement

To develop an AI-based Hospital Discharge Readiness Intelligence Engine that combines structured hospital data and clinical text to estimate operational discharge readiness and identify potential workflow blockers, while providing explainable and human-readable insights through a Gemini-powered chatbot.

1.1 Key Challenges

- Discharge readiness is not represented by a single field.
- Relevant information is distributed across multiple hospital data sources.
- Clinical text contains useful workflow-related information that is difficult to process manually at scale.
- Hospital event tables can contain repeated records and different temporal granularities.
- Missing data and class imbalance can affect machine-learning performance.
- Predicted probabilities may require calibration before being interpreted as readiness estimates.
- Model predictions need explanations that are understandable to non-technical users.
- Clinical data creates a strong requirement for privacy, responsible use, and prevention of information leakage.

2. Proposed Solution

The proposed solution creates a hospitalization-level analytical dataset and combines structured features with information extracted from clinical text. A machine-learning classifier estimates an operational readiness probability. Explainable AI methods such as SHAP are then used to identify important contributing features and potential workflow signals. Finally, a Gemini API chatbot converts the structured model and explanation outputs into concise natural-language responses.

2.1 System Architecture

Architecture Layer	Purpose
MIMIC-IV Patient Records	Source hospital and patient information
Data Preprocessing	Cleaning, datetime conversion, missing-data handling and validation
Structured Features	Hospitalization, laboratory, medication and administrative indicators
Clinical Text → NLP	Text preprocessing and workflow-related signal extraction
Feature Fusion	Combine structured and text-derived features
ML Classification	Estimate operational readiness probability
Calibration + SHAP	Improve probability interpretation and explain predictions
Potential Workflow Blockers	Summarize model-supported operational signals
Gemini AI Chatbot	Explain supplied model outputs in natural language
Streamlit Dashboard	Interactive visualization and user interface

Conceptual flow: MIMIC-IV → preprocessing → structured features + clinical-text NLP → feature fusion → ML model → operational readiness probability → calibration and SHAP → potential workflow blockers → Gemini explanation → Streamlit dashboard.

3. Objectives

1. Collect and preprocess relevant MIMIC-IV hospital data.
2. Create a hospitalization-level dataset using hadm_id as the primary analytical unit.
3. Engineer operationally relevant structured features.
4. Apply NLP to available clinical text and extract workflow-related signals.

5. Compare structured-only, NLP-only, and combined feature models.
6. Train and evaluate suitable classification algorithms.
7. Handle missing values and class imbalance appropriately.
8. Evaluate and calibrate predicted readiness probabilities.
9. Use SHAP to explain individual model predictions.
10. Identify potential workflow blockers from model-supported evidence.
11. Integrate a Gemini API chatbot for human-readable explanations.
12. Build a Streamlit dashboard for interactive analysis.
13. Maintain a clear boundary between operational decision support and clinical decision making.

4. Methodology

4.1 Data Preprocessing

- Load compressed MIMIC CSV files directly using pandas.
- Validate subject_id and hadm_id relationships.
- Convert date/time fields to appropriate datetime types.
- Aggregate event-level records to hospitalization level where required.
- Handle missing values using feature-appropriate strategies.
- Check duplicate and implausible records without altering raw source data.

4.2 Feature Engineering

Potential structured features include length of stay, recent laboratory activity, counts of selected hospital events, medication-related workflow indicators, transfers, diagnosis/procedure indicators, and other operational variables that are available before the prediction cutoff.

4.3 NLP Pipeline

Clinical notes, where authorized, will undergo preprocessing followed by feature extraction. A baseline approach may use TF-IDF features, while more advanced approaches may use clinical embeddings or transformer-based representations if computationally and methodologically justified.

- Text cleaning and normalization
- Tokenization and sentence processing
- Workflow-related entity/status extraction
- TF-IDF or embedding-based representation
- Aggregation of note-level information at hospitalization level

4.4 Machine Learning

Candidate models include Logistic Regression as an interpretable baseline, Random Forest as a tree-based ensemble, and XGBoost or LightGBM as advanced gradient-boosting models. The project will compare structured-only, NLP-only, and fused feature sets.

4.5 Evaluation and Class Imbalance

Model evaluation will use metrics appropriate for classification and potentially imbalanced classes, including precision, recall, F1-score, ROC-AUC, PR-AUC, and the confusion matrix. Class weights or other justified imbalance-handling techniques may be evaluated.

4.6 Probability Calibration

Because the main output is an operational readiness probability, calibration will be evaluated rather than relying only on classification accuracy. Candidate methods include Platt scaling and isotonic regression. Brier score and calibration curves will be used to assess probability quality.

4.7 Explainable AI

SHAP will be used to identify important features contributing to individual predictions. The system will distinguish between model-derived evidence and human interpretation, avoiding unsupported claims about patient safety or clinical status.

5. Gemini AI Chatbot Integration

A Gemini-powered chatbot will be integrated as an explanation and interaction layer. Google’s current Gemini API documentation provides an official Python SDK, installed using the google-genai package, and supports API-key authentication for requests. `!pip install google-genai`

The chatbot will receive controlled context containing the operational readiness score, relevant model explanations, and identified workflow signals. It will answer user questions using only the supplied context and will be instructed not to diagnose, prescribe treatment, or independently decide whether a patient should be discharged.

5.1 Chatbot Interaction Flow

Stage	Description
User question	Natural-language question about a prediction or blocker
Hospitalization context	Selected model output and permitted supporting features
ML + XAI results	Readiness probability, SHAP contributions and potential workflow signals
Gemini API	Generate explanation from controlled context
Response	Concise, professional and uncertainty-aware explanation

5.2 Example Questions

- Why is the estimated operational readiness low?
- What are the main potential workflow blockers?
- Which features contributed most to this prediction?
- Explain the readiness score in simple terms.
- What information is missing for this hospitalization?

6. Streamlit Application

The final application will use Streamlit to present the analytical results through an interactive interface. The dashboard will focus on operational monitoring and model interpretation.

- Overview: total hospitalizations, readiness distribution and summary indicators.
- Hospitalization Analysis: readiness probability, category, contributing features and potential blockers.
- NLP/Text Analysis: extracted workflow-related signals from permitted notes.
- Explainability: SHAP-based feature contributions.
- AI Assistant: Gemini-powered conversational explanation.

7. Expected Outcomes

- A cleaned hospitalization-level analytical dataset.
- An operational discharge-readiness prediction model.
- Comparison of structured, NLP, and fused feature approaches.
- Calibrated readiness probabilities.
- Explainable individual predictions using SHAP.
- Identification of potential workflow blockers supported by available data.
- A Gemini-powered AI assistant for natural-language explanation.
- An interactive Streamlit dashboard demonstrating the complete pipeline.